Set Theory & Computability

A Complete Exam Preparation Guide

Table of Contents

1. Introduction to Sets
2. Set Operations
3. Cardinality
4. Russell's Paradox
5. Countability: $\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$
6. Cantor's Diagonalization Argument
7. Computability and Uncomputability

Topics: Logic · Sets · Cardinality · Computability

Definition: Set

A **set** is an unordered collection of distinct objects, called **elements** or **members**. If x is an element of set A, we write $x \in A$. If not, we write $x \notin A$.

1.1 Notation & Special Sets

Symbol	Name	Description
$\emptyset$ or $\{\}$	Empty Set	Contains no elements
$\mathbb{N}$	Natural Numbers	$\{1, 2, 3, \ldots\}$
$\mathbb{W}$	Whole Numbers	$\{0, 1, 2, 3, \ldots\}$
$\mathbb{Z}$	Integers	$\{\ldots, -2, -1, 0, 1, 2, \ldots\}$
$\mathbb{Q}$	Rationals	$\{p/q \mid p, q \in \mathbb{Z},\ q \neq 0\}$
$\mathbb{R}$	Reals	All points on the number line

Example: Set Descriptions

Sets can be described in three ways:

- **Roster (List):** $A = \{1, 2, 3, 4\}$
- **Set-Builder:** $A = \{x \mid x \in \mathbb{Z} \text{ and } 1 \leq x \leq 4\}$
- **Interval (for $\mathbb{R}$):** $A = [1, 4]$

1.2 Subset and Equality

Definition: Subset

$A \subseteq B$ (A is a **subset** of B) if every element of A is also in B:

$$A \subseteq B \iff \forall x\,(x \in A \Rightarrow x \in B)$$

$A \subset B$ (A is a **proper subset**) means $A \subseteq B$ and $A \neq B$.

Definition: Set Equality

Two sets are equal iff they contain exactly the same elements:

$$A = B \iff (A \subseteq B \text{ and } B \subseteq A)$$

Definition: Power Set

The **power set** of A, written $\mathcal{P}(A)$, is the set of all subsets of A:

$$\mathcal{P}(A) = \{S \mid S \subseteq A\}$$

If $|A| = n$, then $|\mathcal{P}(A)| = 2^n$.

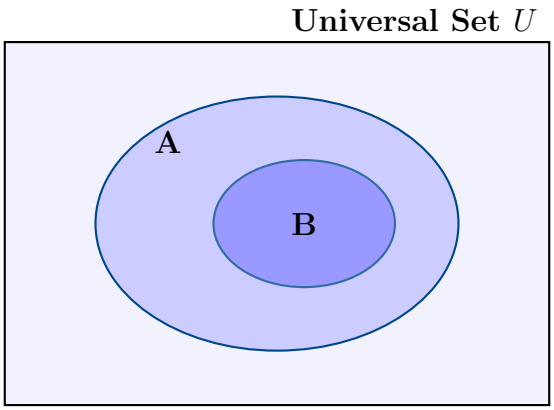
Example: Power Set

Let $A = \{1, 2, 3\}$. Then:

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

$|A| = 3 \Rightarrow |\mathcal{P}(A)| = 2^3 = 8$ ✓

1.3 Visual: Subset Relationships



$B \subset A \subset U$

★ Key Exam Point: Subset vs Element

$\{1\} \subseteq \{1, 2, 3\}$ (subset) vs $1 \in \{1, 2, 3\}$ (element)
$\emptyset \subseteq A$ for ANY set A, but $\emptyset \in A$ only if A explicitly contains $\emptyset$.

Definition: Union

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

Definition: Intersection

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

Definition: Difference & Complement

$$A \setminus B = A - B = \{x \mid x \in A \text{ and } x \notin B\}$$
$$\overline{A} = A^c = U \setminus A = \{x \in U \mid x \notin A\}$$

Definition: Symmetric Difference

$$A \triangle B = (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$$

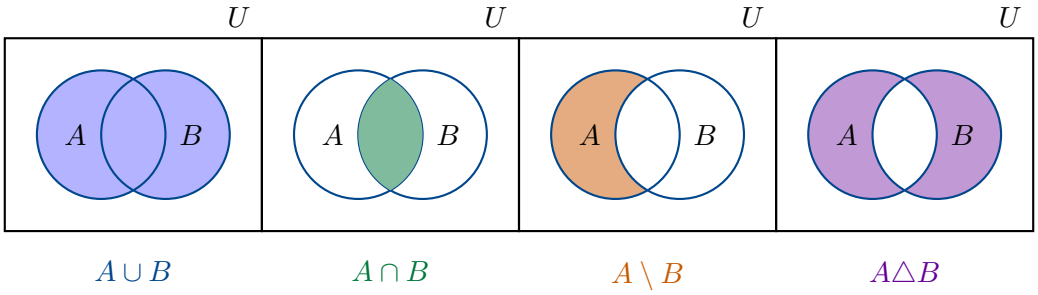
Elements in exactly one of A or B.

Definition: Cartesian Product

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

$|A \times B| = |A| \cdot |B|$

2.1 Venn Diagram Illustrations



2.2 Key Laws (De Morgan's & Others)

Theorem: De Morgan's Laws

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$
$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

"NOT (A or B) = NOT A AND NOT B" *"NOT (A and B) = NOT A OR NOT B"*

Important Set Identities

$A \cup A = A$	$A \cap A = A$	(Idempotent)
$A \cup \emptyset = A$	$A \cap \emptyset = \emptyset$	(Identity/Annihilation)
$A \cup U = U$	$A \cap U = A$	(Domination)
$A \cup B = B \cup A$	$A \cap B = B \cap A$	(Commutative)
$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$	(Associative)
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	(Distributive)

★ Key Exam Point: Exam Tip — Prove Set Equality

To prove $A = B$: Show $A \subseteq B$ (take arbitrary $x \in A$, show $x \in B$) AND $B \subseteq A$. OR use set identities. OR use a Venn diagram for intuition, but algebraic proof is rigorous.

Definition: Cardinality

The **cardinality** of a set A, written $|A|$ or $\#A$, is the number of elements in A.

- A is **finite** if $|A| = n$ for some $n \in \mathbb{W}$
- A is **infinite** if it is not finite

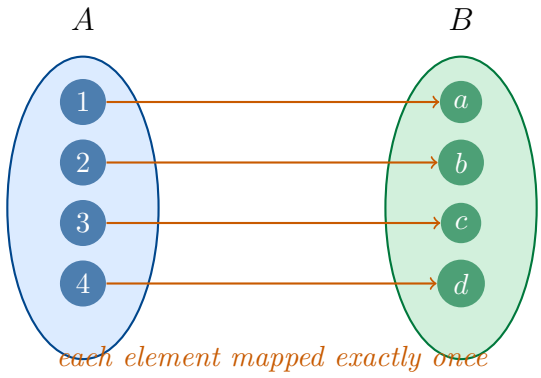
Example: Cardinality Examples

- $|\emptyset| = 0$
- $|\{a, b, c\}| = 3$
- $|\{1, 2, \ldots, 100\}| = 100$
- $|\mathcal{P}(\{a, b, c\})| = 2^3 = 8$
- $|\mathbb{N}| = \aleph_0$ (aleph-null — countably infinite)
- $|\mathbb{R}| = \mathfrak{c}$ (the cardinality of the continuum — uncountably infinite)

Definition: Bijection and Equal Cardinality

Two sets A and B have the same cardinality ($|A| = |B|$) iff there exists a **bijection** (one-to-one and onto function) $f : A \to B$.

Bijection $f : A \to B$



Theorem: Pigeonhole Principle

If n objects are placed in k containers and $n > k$, then at least one container has more than one object.

Russell's Paradox (1901)

Consider the set R defined as:

$$R = \{x \mid x \notin x\}$$

i.e., "the set of all sets that do NOT contain themselves."
Question: Does $R \in R$?

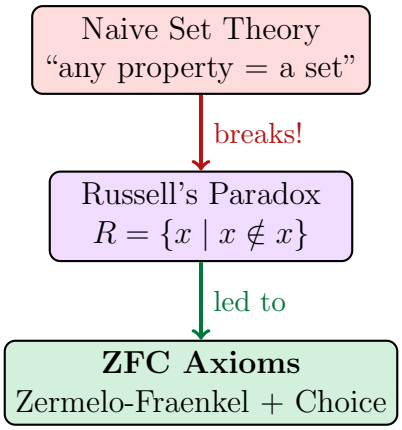
- **If $R \in R$:** Then R satisfies its own condition, so $R \notin R$. **(!!)**
- **If $R \notin R$:** Then R satisfies the condition $x \notin x$, so $R \in R$. **(!!)**

Contradiction either way! This shows that "the set of all sets" leads to a logical inconsistency.

4.1 Implications and Resolution

Note: Why This Matters

Russell's Paradox revealed that **naive set theory** (any property defines a set) is **inconsistent**. This led to the development of formal axiomatic set theories.



Theorem: Resolution via ZFC

In **ZFC set theory**, sets can only be formed from existing sets using restricted comprehension:

$$\{x \in A \mid P(x)\}$$

This prevents self-referential constructions like R and avoids the paradox.

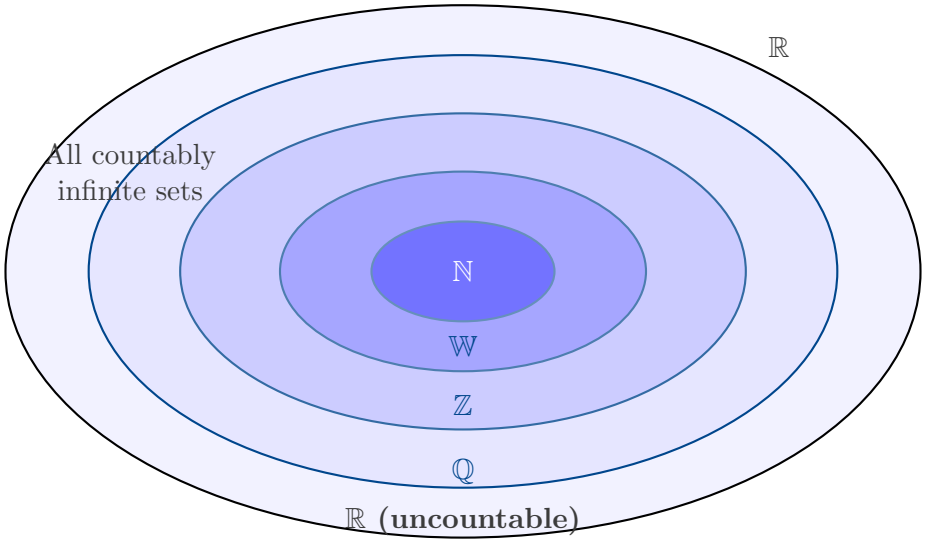
★ Key Exam Point: The Barber Analogy

"A barber shaves all men who do not shave themselves. Who shaves the barber?" This is Russell's Paradox in everyday language — the barber cannot exist just as R cannot exist.

Definition: Countable and Uncountable Sets

- A set S is **countably infinite** if $|S| = |\mathbb{N}|$, i.e., there is a bijection $f : \mathbb{N} \to S$.
- A set is **countable** if it is finite or countably infinite.
- A set is **uncountable** if it is infinite and not countably infinite.

5.1 The Hierarchy of Infinite Sets



5.2 $\mathbb{N}, \mathbb{W}, \mathbb{Z}$ are Countably Infinite

Theorem: $|\mathbb{N}| = |\mathbb{W}| = |\mathbb{Z}|$

All three are countably infinite (cardinality $\aleph_0$).

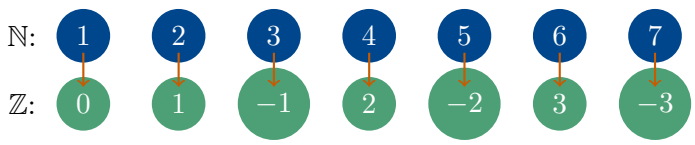
Example: Bijection $\mathbb{N} \to \mathbb{Z}$

The integers can be listed without missing any:

$$0,\ 1,\ -1,\ 2,\ -2,\ 3,\ -3,\ \ldots$$

Formally: $f(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ -(n+1)/2 & \text{if } n \text{ is odd} \end{cases}$

Matching $\mathbb{N}$ to $\mathbb{Z}$



5.3 $\mathbb{Q}$ is Countably Infinite

Theorem: $|\mathbb{Q}| = \aleph_0$

The rationals are countably infinite, even though they are **dense** (between any two rationals lies another).

Example: Cantor's Zig-Zag / Table Argument

Arrange all positive rationals p/q in a grid and traverse diagonally:

Counting $\mathbb{Q}^+$ by Diagonals

$p=1$	$\frac{1}{1}$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$
$p=2$	$\frac{2}{1}$	$\frac{2}{2}$	$\frac{2}{3}$	$\frac{2}{4}$	$\frac{2}{5}$
$p=3$	$\frac{3}{1}$	$\frac{3}{2}$	$\frac{3}{3}$	$\frac{3}{4}$	$\frac{3}{5}$
$p=4$	$\frac{4}{1}$	$\frac{4}{2}$	$\frac{4}{3}$	$\frac{4}{4}$	$\frac{4}{5}$
$p=5$	$\frac{5}{1}$	$\frac{5}{2}$	$\frac{5}{3}$	$\frac{5}{4}$	$\frac{5}{5}$
	$q=1$	$q=2$	$q=3$	$q=4$	$q=5$

Zig-zag path covers all p/q

5.4 $\mathbb{R}$ is Uncountable

Theorem: $|\mathbb{R}| > \aleph_0$

The real numbers are **uncountably infinite**. Their cardinality $\mathfrak{c} = |\mathbb{R}| = 2^{\aleph_0} > \aleph_0$.

Note: Proof Strategy

This is proved using **Cantor's Diagonalization Argument** — see Section 6!

★ Key Exam Point: Summary Table for Exams

Set	Cardinality	Type	Bijection with $\mathbb{N}$?
$\mathbb{N}$	$\aleph_0$	Countably ∞	Identity
$\mathbb{W}$	$\aleph_0$	Countably ∞	$f(n) = n - 1$
$\mathbb{Z}$	$\aleph_0$	Countably ∞	Interleaving
$\mathbb{Q}$	$\aleph_0$	Countably ∞	Diagonal
$\mathbb{R}$	$\mathfrak{c} = 2^{\aleph_0}$	**Uncountable**	**None exists!**

The Big Idea

Cantor's diagonalization argument proves that **no list (bijection from $\mathbb{N}$) can cover all real numbers** in $[0, 1]$. Therefore $\mathbb{R}$ is uncountable.

6.1 The Proof

Theorem: Cantor's Theorem: $[0,1]$ is Uncountable

Proof by contradiction. Suppose $[0,1]$ *is* countable. Then we can list all reals as:

$$\begin{aligned} r_1 &= 0.\ \mathbf{d_{11}}\ d_{12}\ d_{13}\ d_{14}\ \cdots \\ r_2 &= 0.\ d_{21}\ \mathbf{d_{22}}\ d_{23}\ d_{24}\ \cdots \\ r_3 &= 0.\ d_{31}\ d_{32}\ \mathbf{d_{33}}\ d_{34}\ \cdots \\ r_4 &= 0.\ d_{41}\ d_{42}\ d_{43}\ \mathbf{d_{44}}\ \cdots \\ &\vdots \end{aligned}$$

Construct a new number $s = 0.s_1 s_2 s_3 s_4 \cdots$ where:

$$s_i = \begin{cases} 5 & \text{if } d_{ii} \neq 5 \\ 6 & \text{if } d_{ii} = 5 \end{cases}$$

Then s differs from r_i in the i-th decimal place for *every* i. So $s \notin \{r_1, r_2, r_3, \ldots\}$. **(!!)**
Contradiction: the list was supposed to contain ALL reals. Hence $[0,1]$ is uncountable. ■

6.2 Visual: The Diagonal Construction

Cantor's Diagonalization: Change the Diagonal!

	$d_{1,1}$	$d_{2,2}$	$d_{3,3}$	$d_{4,4}$	$\cdots$
$r_1 = 0.$	**4**	1	7	3	$\cdots$
$r_2 = 0.$	9	**2**	8	1	$\cdots$
$r_3 = 0.$	3	6	**5**	9	$\cdots$
$r_4 = 0.$	0,4,2,7				$\cdots$
$\vdots$					
	6	**7**	**6**	**8**	$\cdots$

Change each diagonal digit

s differs from every r_i — it's not on the list!

★ Key Exam Point: Key Takeaways from Diagonalization

- $|\mathbb{N}| = \aleph_0$ (countably infinite)
- $|\mathbb{R}| = 2^{\aleph_0} = \mathfrak{c}$ (uncountably infinite, strictly larger!)
- The method: build a number that *disagrees* with every element of a supposed

list

- Same technique proves the *Halting Problem is undecidable* (Section 7)

Note: Cantor's Theorem (General)

For any set A: $|A| < |\mathcal{P}(A)|$. There is no largest cardinality — the hierarchy of infinities never ends:

$$\aleph_0 < 2^{\aleph_0} < 2^{2^{\aleph_0}} < \cdots$$

Definition: Algorithm & Computable Function

An **algorithm** is a finite, step-by-step procedure that terminates and produces an output. A function f is **computable** if there exists an algorithm (Turing Machine) that computes $f(x)$ for every input x.

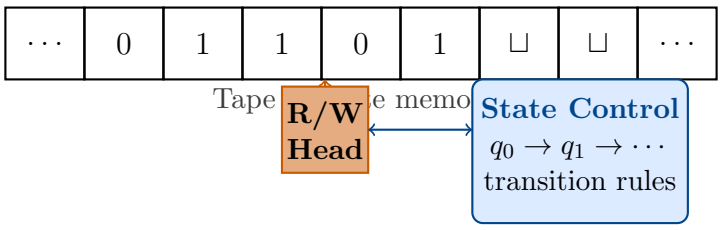
Definition: Turing Machine (TM)

A **Turing Machine** is a theoretical model of computation consisting of:

- An infinite tape (memory)
- A read/write head
- A finite set of states and transition rules

Turing Machines define the limits of what is **algorithmically computable**.

Turing Machine Model



7.1 The Halting Problem

The Halting Problem

Question: Does there exist an algorithm H that, given any program P and input x, decides whether P eventually **halts** (stops) or runs **forever**?

$$H(P,x) = \begin{cases} \texttt{HALT} & \text{if } P(x) \text{ halts} \\ \texttt{LOOP} & \text{if } P(x) \text{ loops forever} \end{cases}$$

Answer: NO! The Halting Problem is **undecidable**.

Theorem: Halting Problem is Undecidable (Turing, 1936)

Proof by contradiction (diagonalization).
Suppose H exists. Construct program D (the "Diagonal" program):

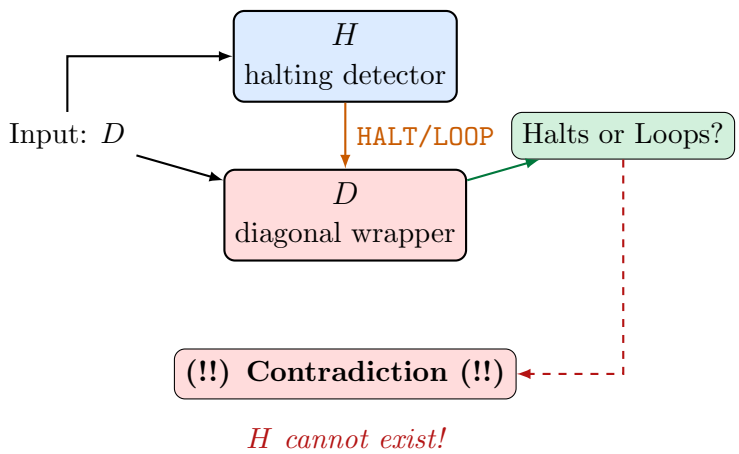
$$D(P) = \begin{cases} \text{loop forever} & \text{if } H(P,P) = \texttt{HALT} \\ \text{halt} & \text{if } H(P,P) = \texttt{LOOP} \end{cases}$$

Now ask: what does $D(D)$ do?

- If $D(D)$ halts: then $H(D,D) = \texttt{HALT}$, so $D(D)$ loops forever. **(!!)**
- If $D(D)$ loops: then $H(D,D) = \texttt{LOOP}$, so $D(D)$ halts. **(!!)**

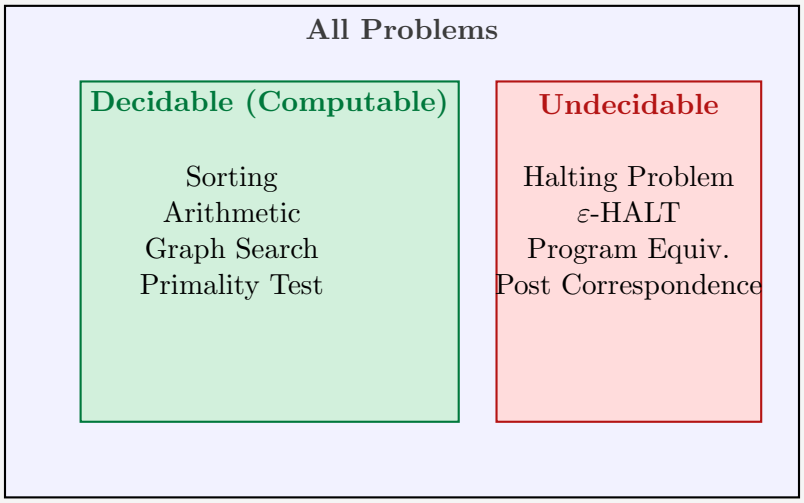
Contradiction in both cases. Therefore H cannot exist. ■

The Diagonal Trick for the Halting Problem



7.2 Computable vs Uncomputable

Landscape of Problems



7.3 Recognizable vs Decidable

Definition: Turing-Recognizable

A language L is **Turing-recognizable** (semi-decidable) if there exists a TM that *accepts* all strings in L but may *loop forever* on strings not in L.

Definition: Decidable

A language L is **decidable** if there exists a TM that *always halts* and correctly accepts or rejects every input.

Hierarchy of Languages

★ Key Exam Point: Connection: Diagonalization → Halting Problem

Cantor's diagonalization (Section 6) is the same logical structure as Turing's proof:

1. Assume you have a complete list / deciding machine
2. Construct a new object that differs from every entry on the diagonal
3. Contradiction — the list / machine is incomplete / impossible

Both use the same fundamental trick!

7.4 Rice's Theorem

Theorem: Rice's Theorem

Any **non-trivial** property of the language recognized by a Turing Machine is **undecidable**.

Note: What This Means Practically

You cannot write a general program that looks at another program and decides any interesting behavioral property (does it output 42? does it accept all inputs? etc.) — all such questions are undecidable.

Master Summary for Exams

1. Sets: Unordered, distinct. $\emptyset \subseteq$ everything. $|\mathcal{P}(A)| = 2^{|A|}$.
2. Operations: $\cup, \cap, \setminus, \triangle, \times$. De Morgan: $\overline{A \cup B} = \bar{A} \cap \bar{B}$.
3. Cardinality: $|A| = |B|$ iff bijection $A \to B$ exists. Pigeonhole for finite sets.
4. Russell's Paradox: $R = \{x \mid x \notin x\}$ is self-contradictory $\Rightarrow$ ZFC axioms needed.
5. Countability: $|\mathbb{N}| = |\mathbb{W}| = |\mathbb{Z}| = |\mathbb{Q}| = \aleph_0$; $|\mathbb{R}| = \mathfrak{c} > \aleph_0$.
6. Diagonalization: Cantor proves $\mathbb{R}$ uncountable by constructing a number that differs from every listed real on the diagonal. Key lemma: list must be incomplete.
7. Computability: Decidable $\subsetneq$ Recognizable $\subsetneq$ All Languages. Halting Problem is undecidable (proof by diagonalization). Rice's Theorem: no non-trivial program property is decidable.